

HR Attrition Cleaning Report

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Dataset: HR Dataset
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GitHub: <https://github.com/Mpho048/Data-Cleaning>

Dataset: HR Dataset

This report contains data
cleaning results and dataset
information.

Generated automatically using
Python and FPDF.

Dataset Preview:

employee_id	age	department	monthly_income	attrition
EMP41483	45	Sales	14327	No
EMP23354	34	Human Resources	17588	No
EMP14171	53	Marketing	6751	No
EMP33840	18	Human Resources	10939	No
EMP15651	46	IT	6110	No
EMP03397	59	Operations	19727	No

Dataset Information

Rows	Columns	Missing	Memory Usage
50020	20	5	28.76 MB

Column	Data Type	Missing
employee_id	str	2
age	int64	0
department	str	1
job_role	str	0
gender	str	0
monthly_income	int64	0
job_level	int64	0
years_at_company	int64	0
years_in_role	int64	0
education_field	str	0
business_travel	str	1
marital_status	str	0
work_life_balance	int64	0
job_satisfaction	int64	0
performance_rating	int64	0
overtime	str	0
num_companies_worked	int64	0
training_times_last_year	int64	0
attrition	str	1
attrition_cost	int64	0

Data Cleaning Summary

- 1. Removed duplicate records.**
- 2. Shape of the data after cleaning, rows: 49995 and columns: 20.**
- 3. Trimmed leading and trailing whitespace from text columns.**
- 4. Standardized department names.**
- 5. Corrected inconsistent attrition values.**
- 6. Replaced invalid department entries.**
- 7. Filled missing values where appropriate.**
- 8. Removed records with missing employee IDs or attrition status.**
- 9. Corrected invalid age values using the median age.**
- 10. Converted negative numeric values to positive values where applicable.**
- 11. Investigated potential outliers in monthly income and only 4 where detected**